Modelling Moon Phases · Tier Pack · Independent

Learning goal: I can test a 3D model of the Moon's orbit to record all eight phases and evaluate the model.

Independent Route (Book First)

Complete the 8-position Moon phase diagram (shading the dark part at each position), label the phases, and state one strength and one limitation of the ball-and-torch model.

Help: From Pos 1 to Pos 5 the lit part grows (__); from Pos 5 to Pos 1 it shrinks (__). A strength of the model is __, while a limitation is __.

Optional Make It · Independent

Make an 8-Phase Moon Cycle Flip Strip

Shade and label all 8 phases in sequence from Position 1 (New Moon) to Position 8 (Waning Crescent) on a strip, and write your model evaluation in your book.

Word Help

waxing crescent: the phase after New Moon when a thin sliver of light grows on the right

waxing gibbous: the phase before Full Moon when more than half of the face is lit and growing

waning gibbous: the phase after Full Moon when more than half is lit and shrinking

waning crescent: the final thin sliver of light before

the cycle returns to New Moon

Exit Check

In our model, what did the torch, the observer, and the ball represent?

Write the four main phases in order starting from New Moon.